



IRRIGATION AND CANALS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Pattiseema Project — Godavari-Krishna Link

- Pattiseema Lift Irrigation Project Godavari river ko Krishna river se jodta hai (river interlinking project). [Q1]

2. Canal Project-State Match (Jawai-Rajasthan Trap)

- Ghatprabha Canal — Karnataka, Ukai Canal — Gujarat (Tapti river par), Nira Canal — Maharashtra (Bhima ki tributary Nira river par) sahi hain — Jawai Project 'Tamil Nadu' mein NAHI hai, ye Rajasthan mein hai (Luni ki tributary Jawai river par). [Q2]



EXAM TRAP — Jawai Project — Rajasthan mein hai (Luni ki tributary), Tamil Nadu mein NAHI

3. Peninsular India — Pond Irrigation (Assertion-Reason)

- Peninsular India mein **talab (pond) irrigation ka major source hai** (Assertion sahi) kyunki **undulating relief aur hard rock canals/wells khodna mushkil banati hai, aur zyadatar rivers seasonal hain (summer mein sookh jaati hain)** (Reason sahi explanation hai). [Q3]

4. Dominant Irrigation Source — Tubewells (Not Wells)

- 2021-22 data ke mutabik India mein irrigation ka main source **Tubewells (47.3%)** hai — 'Wells' galat **statement hai** (tubewell+wells combined 60.5%, canals 24.7%, tanks 2.8%). [Q4]

5. State-wise Irrigation Percentage (Haryana Highest)

- 2011 data ke mutabik cropped-area-under-irrigation % **Haryana (83.90%)** **sabse zyada hai** — UP (72.76%), MP (28.20%), Maharashtra (16.78%) se zyada. **2015-16 Agriculture Census ke mutabik Chandigarh (100%)** **sabse zyada hai, Punjab (99.6%)** **dusre number par.** [Q5, Q6]

6. Micro-Irrigation Benefits

- Micro-irrigation ke sahi benefits: **fertilizer/nutrient loss kam hoti hai, groundwater level ka receding rukta hai** — 'dry land farming mein ekmatra means' **GALAT hai.** [Q7]

7. Protective Irrigation — P.W.P. Concept

- Life-saving/Protective irrigation **P.W.P. (Permanent Wilting Point)** **par irrigation ko kehte hain** — is point par plants ke liye moisture insufficient ho jaati hai, irrigation compulsory ban jaata hai. [Q8]

8. Tubewell Irrigation Development (Saryupar Plain, UP)

- Pichhle 25 saalon mein tubewell irrigation ka **sabse spectacular development Saryupar Plain (canals ki kami ki wajah se)** mein hua hai — Ganga-Ghaghra Doab, Ganga-Yamuna Doab, Bundelkhand se zyada. [Q9]
- India mein **Uttar Pradesh mein maximum tubewell-irrigated area (73.9%)** hai — **UP tubewell+well irrigation mein bhi sabse aage hai (10299 thousand hectares tubewell se, 1269 thousand hectares wells se).** [Q10, Q11, Q12]

9. Groundwater Resources Ranking (UP Highest)

- Irrigation ke liye available groundwater resources descending order: **Uttar Pradesh (41.29 BCM) > Maharashtra (15.29 BCM) > West Bengal (10.84 BCM) > Assam (1.97 BCM).** [Q13]

10. Garland Canal System & River Interlinking Pioneers

- Garland Canal System **Captain Dineshaw J. Dastur ne propose kiya tha** — River interlinking ka concept **sabse pehle M. Visvesvaraya ne diya, baad mein K.L. Rao ne bhi ek alag project propose ki — dono projects reject kar di gayi thi.** [Q14]

11. Sir Arthur Cotton — South India's Irrigation Pioneer

- South India mein irrigation works ka pioneer **Sir Arthur Cotton** hai — British irrigation engineer, **poore British India mein irrigation-navigation canals banwaye.** [Q15]

12. Irrigation Project Classification (Minor/Medium/Major)

- Projects classification: **Minor** (CCA ≤ 2000 hectares), **Medium** (2000-10000 hectares), **Major** (>10000 hectares) — Minor projects se India ki 62% irrigation potential banti hai (Major+Medium se 38%). [Q16]

13. Farakka Canal — Water Capacity & Purpose

- Farakka Canal (Murshidabad, WB, 1975 mein bana) ki water-carrying capacity **40,000 cusecs** hai — Kolkata Port (ab Syama Prasad Mookerjee Port) ki navigability improve karne ke liye banayi gayi thi. [Q17]

14. Mangalam Project — Kerala

- Mangalam Irrigation Project **Kerala** (Palakkad district, Bharathapuzha river system) mein hai. [Q18]

15. Saran Canal — Gandak River Origin

- Saran Irrigation Canal **Gandak river** (Balmiki Nagar barrage) se nikaali gayi hai. [Q19]

16. Indira Gandhi Canal — Harike Barrage & Rivers

- Indira Gandhi Canal **Harike Barrage** (Firozpur, Punjab) — Satluj aur Beas ke sangam ke paas se nikalti hai — length 649km, Sri Ganganagar, Bikaner, Jodhpur, Jaisalmer irrigate karti hai. Water supply Satluj, Beas aur Ravi teeno se milti hai (Chenab se NAHI). [Q20, Q21, Q22, Q23, Q24]

17. Beas Project — Pong Barrage

- Beas Project (Punjab-Haryana-Rajasthan joint) **Pong barrage** se Indira Gandhi Canal ko winter mein water supply maintain karta hai. [Q25]

18. Gang Canal — Oldest Canal System (1927)

- Gang Canal, duniya ke sabse achhe-developed aur puraane canal systems mein se ek hai — **Maharaja Ganga Singh** (Bikaner) ne 1927 mein banwaya, Hussainiwala (Firozpur) se Satluj river se nikalti hai. [Q26, Q28]

19. Canal-River Match Set (Upper Bari Doab, Sirhind, Gurgaon)

- Canal-river match: **Upper Bari Doab Canal** — Ravi, **Sirhind Canal** — Sutlej, **Gurgaon Canal** — Yamuna, **Indira Gandhi Canal** — Sutlej+Ravi+Beas. [Q27]

20. Statement Verification — Nizam Sagar, Narmada, Penganga

- Sahi statements: **Narmada Amarkantak** se nikalti hai, **Nizam Sagar** Manjra river (Godavari tributary) par hai, **Penganga** Godavari ki tributary hai — 'Ghaggar ka paani Indira Gandhi Canal mein use hota hai' **GALAT** hai (Sutlej-Beas-Ravi se hi paani aata hai, Ghaggar se nahi). [Q29]

21. Lower Ganga Canal — Narora Origin

- Lower Ganga Canal **Narora** (Bulandshahr) se Ganga river se nikalti hai — 1878 mein complete hui thi. [Q30]

22. Bihar's Triveni Canal

- Bihar (north-western part) ki irrigation **Triveni Canal** se hoti hai — Upper Ganga, Sharda, Eastern Yamuna Canal se nahi. [Q31]
- Sharda Tributary Command Development Project ke objectives: **agricultural production badhaana, multiple-crop cultivation se land-use format change karna, land management reforms** — teeno sahi hain (CADP ke andar Sharda, Ramganga, Gandak — 3 projects shuru hue the). [Q32]

23. Hariyali Yojana — Watershed/Water Management Scheme

- Hariyali scheme **wasteland development** ki nayi scheme hai — **PM Atal Bihari Vajpayee ne 27 Jan 2003 ko watershed management ke tahat launch kiya** — panchayats ke through chalti hai, afforestation ko encourage karti hai. Broadly **water management se related** hai. [Q33, Q34]

24. Watershed Programmes — Ministry Match & IWMP Merger

- Integrated Watershed Development Programme ke benefits: **soil runoff prevention, rainwater harvesting/groundwater recharge, natural vegetation regeneration** — 'perennial rivers ko seasonal rivers se jodna' benefit mein include **NAHI** hai. [Q35]
- 3 watershed programmes (IWDP, DPAP, DDP) **Ministry of Rural Development ke Department of Land Resources** ke through 2008 mein IWMP mein merge hue — **koi bhi diya gaya ministry-match sahi NAHI hai (Agriculture/Environment/Rural Development directly nahi)**, NWDPRA **Ministry of Agriculture** ka initiative hai. [Q36]

25. Drip Irrigation — Advantages

- Drip irrigation ke advantages: **weed mein kami, soil erosion mein kami** — 'soil salinity mein kami' iska **direct advantage NAHI hai** (options mein diya gaya galat combination hai). [Q37]

26. Buckingham Canal — Coromandel Coast Navigation

- Buckingham Canal (freshwater navigation canal, length 796km) **Coromandel Coast** ke parallel hai — Andhra Pradesh ke Kakinada se Tamil Nadu ke Parangipettai (Chidambaram ke paas) tak, Chennai port se backwaters ko jodta hai. [Q38]

27. Water Consumption — Agricultural Sector Highest

- India mein **Agricultural sector mein pani ki khapat sabse zyada** hai — total rainfall se water availability ~3880 BCM/year hai. [Q39]

28. Chambal Canal Project — Yamuna/Ganga Basin Dispute

- Chambal Canal Irrigation Project **Yamuna Basin se related hai (smaller-basin perspective se)** — lekin **Ganga Basin se bhi related hai (larger-basin perspective se, kyunki Chambal-Yamuna dono Ganga mein milti hain)** — isliye ye disputed question hai, dono answers sahi maane jaate hain. [Q40]